PROTOTYPE DESCRIPTION

Smart Safety Companion

Overview

The Smart Safety Companion is a mobile app prototype developed using MIT App Inventor. It helps users during emergencies by detecting strong phone shaking and showing an alert with location and direction details.

1. Purpose of the Prototype

The main aim of this prototype is to improve personal safety. It does the following:

- Monitors phone movement continuously
- Detects sudden or strong shaking
- Tracks the user's GPS location
- Identifies the direction of the device
- Generates emergency alert messages
- Stores emergency contact numbers

This app works as a safety assistant that reacts automatically in risky situations.

2. User Interface Description

Application Title

The title "Smart Safety Companion" is displayed at the top of the screen.

Emergency Contact Section

- Users can enter emergency contact numbers
- Contacts are saved using TinyDB
- Saved numbers remain even after closing the app
- These contacts are used during emergencies

Start Monitoring Button

- Starts the monitoring process
- Activates accelerometer, location, and orientation sensors
- Changes the status to “Monitoring”
- Begins tracking phone movement

Stop Monitoring Button

- Stops all monitoring activities
- Turns off all sensors
- Resets values
- Changes the status to “Stopped”

3. Sensor and Status Display

The app shows real-time data on the screen:

- Status – Shows whether the app is Stopped, Monitoring, or Emergency Detected
- Accelerometer Values (X, Y, Z) – Shows phone movement data
- Shake Value – Calculated as $|X| + |Y| + |Z|$
- Location – Displays latitude and longitude
- Direction – Shows North, East, South, or West
- Alert Message – Displays emergency alert with time, location, and direction

4. Background Design

A technology-based background image is used to make the app look attractive and user-friendly.

5. Block Programming Logic

The app is built using MIT App Inventor blocks.

Global Variables

- monitoring – Checks whether monitoring is ON or OFF
- cooldown – Prevents repeated alerts
- shakeThreshold – Minimum shake value to detect emergency
- directionText – Stores direction information
- alertMessage – Stores alert message

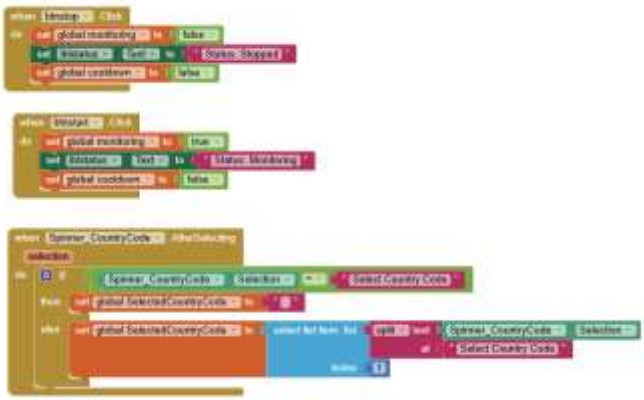
Working of Sensors

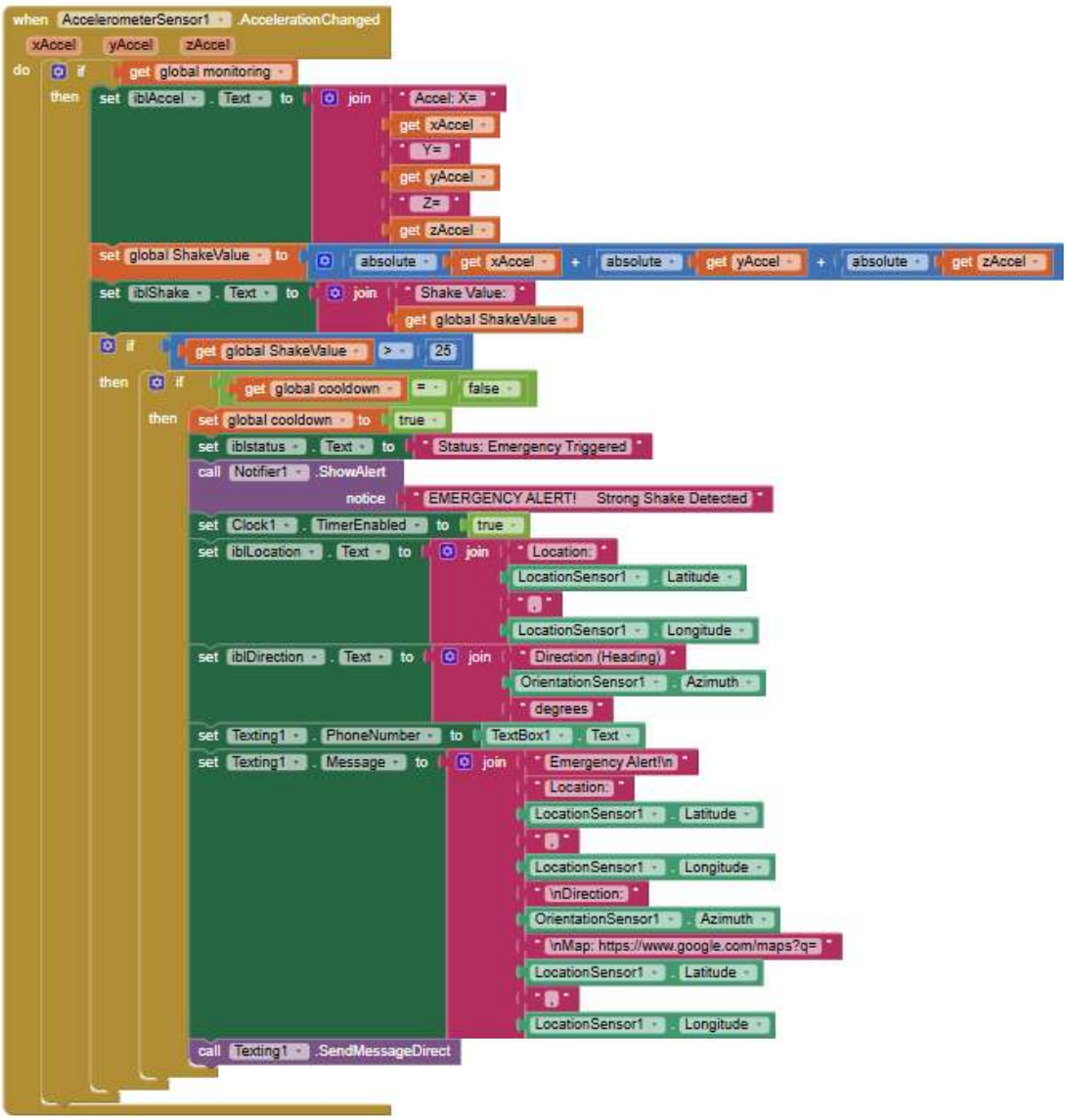
- The accelerometer checks phone movement
- Shake value is calculated using X, Y, and Z values
- If shake value exceeds the threshold:
    - Emergency is detected
    - Location and direction are collected
    - Alert message is generated
    - Cooldown is activated

Direction Detection

Direction is identified using orientation sensor:

- 315° – 45° → North
- 45° – 135° → East
- 135° – 225° → South
- 225° – 315° → West




when AccelerometerSensor1 .AccelerationChanged
xAccel yAccel zAccel
do if get global monitoring
then set lblAccel . Text to join "Accel: X=" get xAccel " Y= " get yAccel " Z= " get zAccel
set global ShakeValue to absolute get xAccel + absolute get yAccel + absolute get zAccel
set lblShake . Text to join "Shake Value: " get global ShakeValue
if get global ShakeValue > 25
then if get global cooldown = false
then set global cooldown to true
set lblstatus . Text to "Status: Emergency Triggered"
call Notifier1 .ShowAlert
notice "EMERGENCY ALERT! Strong Shake Detected"
set Clock1 . TimerEnabled to true
set lblLocation . Text to join "Location: " LocationSensor1 . Latitude ", " LocationSensor1 . Longitude
set lblDirection . Text to join "Direction (Heading) " OrientationSensor1 . Azimuth " degrees"
set Texting1 . PhoneNumber to TextBox1 . Text
set Texting1 . Message to join "Emergency Alert!\n" "Location: " LocationSensor1 . Latitude ", " LocationSensor1 . Longitude "\nDirection: " OrientationSensor1 . Azimuth "\nMap: https://www.google.com/maps?q=" LocationSensor1 . Latitude ", " LocationSensor1 . Longitude
call Texting1 .SendMessageDirect

6. Overall Working

- User enters emergency contacts
- User starts monitoring
- App continuously checks phone movement
- If strong shaking is detected:
    - Emergency alert is shown
    - Location and direction are displayed
- User can stop monitoring at any time

7. Conclusion

The Smart Safety Companion is a simple safety app prototype that uses phone sensors to detect emergencies.
It combines motion detection, location tracking, and direction information to support users during dangerous situations.

This prototype can be extended for:

- Women safety
- Accident detection
- Elderly care
- Emergency alert systems